

OPINION

Mitosis is not a key target of microtubule agents in patient tumors

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Abstract | Mitosis-specific agents have, to date, not been clinically successful. By contrast, microtubule-targeting agents (MTAs) have a long record of success, usually attributed to the induction of mitotic arrest. Indeed, it was this success that led to the search for mitosis-specific inhibitors. We believe the clinical disappointment of mitosis-specific inhibitors stands as evidence that MTAs have been successful not only by interfering with mitosis but, more importantly, by disrupting essential interphase cellular mechanisms. In this Perspective we will review literature that supports a paradigm shift in how we think about one of our most widely used classes of chemotherapeutics—MTAs. We believe that the steady presence and constant physiological role of microtubules are responsible for the overall success of MTAs. While mitosis-specific inhibitors are effective on only a small fraction of the tumor mass (dividing cells), MTAs target tubulin, a protein that has crucial roles in both mitotic and non-mitotic cells.

Komlodi-Pasztor, E. *et al.* *Nat. Rev. Clin. Oncol.* **8**, 244–250 (2011); published online 1 February 2011; corrected online 4 March 2011; doi:10.1038/nrclinonc.2010.228

Introduction

Much of our knowledge of the mechanism of action of microtubule-targeting agents (MTAs) is based on studies of their effect on microtubules in solution, and from studies of rapidly-dividing cells in culture. From studies that assess microtubules polymerized from purified tubulin in solution, it is well established that MTAs act by either disrupting microtubule polymerization (for example, colchicine and vinca alkaloids) or hyper-stabilizing microtubule polymers (for example taxanes and epothilones).¹ From studies on cells in culture it is clear that mitosis is the main target of MTAs in these rapidly-dividing cells.¹ However, microtubules are as important in non-dividing cells as they are in the mitotic spindle of dividing cells. In this Perspective, we discuss the following: first, in rapidly-dividing cells, MTAs target mitosis; second, MTAs cannot target mitosis in cells that are not dividing, or are dividing very slowly, because mitosis is rare or absent; third, the majority of human tumors divide very slowly compared with cultured cell lines and tumors in animal models, and hence mitosis is an unlikely

target in the majority of human tumors; and finally, because MTAs are effective against (some) human tumors, including some very slowly dividing ones, these drugs must target the non-mitotic (that is interphase) functions of microtubules.

We suggest that cellular functions sensitive to MTAs include the trafficking on microtubules that is essential for cell metabolism and/or signaling, and that relies on organized arrays of microtubules. First we briefly review the organization of microtubules in polarized arrays in interphase as well as in mitotic cells. Then we discuss the transition to mitosis and the role of mitosis-specific enzymes. Finally we relate these to tumor growth in xenograft models and in spontaneous human tumors and the effectiveness of mitosis-directed chemotherapy as compared with microtubule-directed chemotherapy.

Microtubules in the cell cycle

Studies in solution with microtubules polymerized from purified tubulin provide valuable information regarding the kinetics of microtubule dynamics and the interaction of a drug with microtubules. However, microtubule functions in cells are dependent on additional organization not found in randomly oriented, independent microtubules

in vitro. This is because independent microtubules are rare in cells; rather, cellular microtubules occur in ordered arrays *in vivo*. In animal cells, microtubule-organizing centers (MTOCs) are the site of microtubule nucleation—the structures from which microtubules emerge (Figure 1).² In animal cells, the two MTOCs associated with spindle formation are called the centrosomes. A major function of the centrosome is the organization of the two opposing arrays that form the mitotic spindle apparatus required to separate chromosomes during cell division.² However, this is not the only function of the MTOC in the life of a cell. In interphase, the MTOC organizes the array of microtubules that comprise the interphase cytoskeleton, which is essential for cellular trafficking, and the formation of cilia and flagella.^{3,4} In most interphase cells, the microtubules are organized into parallel, polarized arrays with their active or ‘plus’ ends oriented in the same direction—for example outward from the nucleus in growing epithelial cells; apical to basal in polarized cells.⁵ Establishment of this polarized array of microtubules in mammalian cells occurs because microtubules do not freely polymerize in the cytoplasm, but only polymerize following nucleation from a structure that initiates all microtubules with their plus ends oriented out from the nucleating center (Figure 1).⁶ This orientation is recognized by motor proteins (dynein and kinesin), which allows for directed movement of cargo on microtubules. Included among the cargo are vesicles containing proteins to be secreted from the cell and components essential for growth, which are carried on the kinesin family of proteins to the cell’s periphery from sites of production located centrally;⁷ and signaling proteins moving to the nucleus on dynein family members.⁸ Thus, while centrosomal nucleation of polarized arrays of microtubules is essential for mitosis, the same function is essential for cellular organization in the large portion of the cell cycle that is not mitosis.

Mitotic kinases

Mitosis is a highly complex and precisely regulated process that ensures the separation of sister chromatids through the tension generated by microtubule action at

Competing interests

The authors declare no competing interests.

the kinetochores.⁹ Spindle fibers attached to kinetochores direct the movement of sister chromatids to opposite spindle poles. However, mitosis requires the participation of many other proteins in addition to tubulin polymerized as microtubules. Many of these proteins are only active during mitosis and are responsible for controlling different steps in the assembly and function of the complex machinery of the mitotic spindle. The mitotic spindle checkpoint ensures the proper attachment of the mitotic spindle to kinetochores and the existence of the tension necessary to separate the sister chromatids. Activated checkpoint proteins bound to the kinetochore form a macromolecular complex and can initiate a signal to prevent the separation of the sister chromatids and the onset of anaphase if attachment and tension are not correct.⁹ The past decade has brought a greater understanding of the complex choreography of mitosis, as proteins tasked with different functions have been identified and characterized.¹⁰ Among these proteins, the aurora kinases and polo-like kinases emerged as targets for cancer therapeutics.^{11–13} Driven by the thesis that MTAs kill cancer cells by interfering with mitosis, many advocated that by specifically targeting the mitotic apparatus, drugs inhibiting mitotic kinases would achieve the same or greater cytotoxicity without the troublesome side effects of peripheral and autonomic neuropathy.^{14–16}

Aurora A localizes to the centrosome during the S-phase of the cell cycle (Figure 2).¹⁷ As the cell cycle progresses through prophase, metaphase, and anaphase, the level of aurora A increases and the protein translocates to the mitotic poles and the adjacent spindle microtubules. During telophase and cytokinesis, aurora A is located in the midbody. Aurora B and aurora C are chromosome passenger proteins.¹⁸ Aurora B localizes to microtubules near kinetochores, especially to K-fibers—bundles of 15 to 30 specialized microtubules that connect kinetochores to spindle pole fibers—while aurora C moves from the centromeres to the midzone of mitotic cells as the cell cycle progresses.¹⁸ The polo-like kinases (PLK1–4) localize to the centrosome during interphase and prophase, move to the spindle poles following prophase, and relocate to the midbody for telophase and cytokinesis (Figure 2).^{19,20} Supplementary Table 1 online summarizes the substrate proteins through which the effects of these mitotic proteins are mediated; they have crucial roles in initiating, advancing and concluding the process of mitosis. They

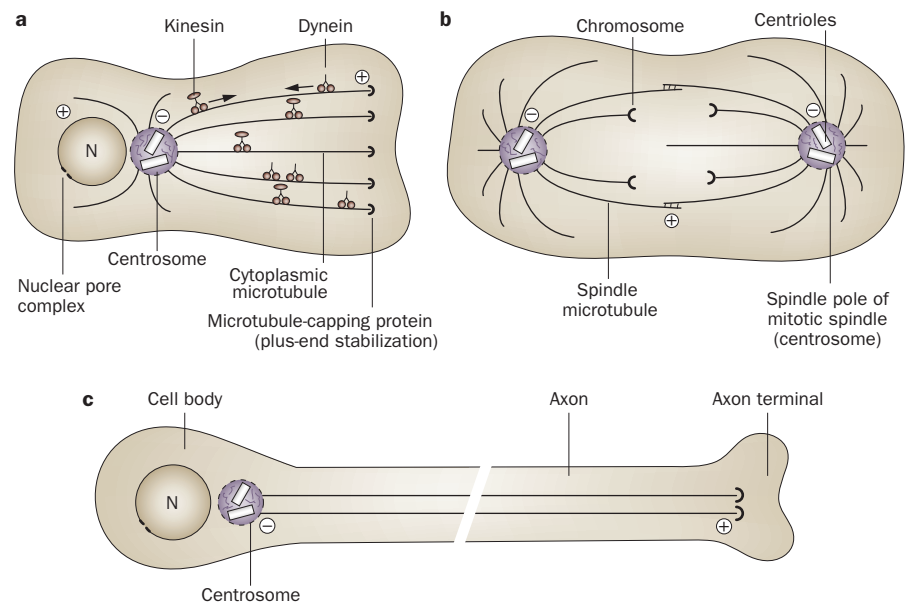


Figure 1 | Cellular organization depicting structures and proteins involved in cell division.

a | Interphase cell. **b** | Dividing (mitotic) cell. **c** | Nerve cell. In a dividing cell the spindle fibers that comprise the mitotic spindle attach to chromosomes at the kinetochores and direct the movement of sister chromatids to opposite spindle poles. Structures depicted are the nuclear pore complex (responsible for the exchange of molecules between the cytoplasm and the nucleus), centrosome (two microtubule organizing centers associated with spindle formation), kinesin (plus-end directed motor protein responsible for the movement of cargo on microtubules towards the cell periphery), dynein (minus-end directed motor protein responsible for the movement of cargo on microtubules towards the nucleus), microtubules, microtubule-capping protein (protein structure at the plus end of the microtubules), axon (long projection of a neuron), and axon terminal (distal termination of an axon, responsible for signal transduction). Permission obtained from L. A. Staehelin, University of Colorado, Boulder, CO, USA.

often act in concert, but seem to perform unique functions, without overlap.

Kinesin spindle protein (KSP), similar to the mitotic kinases, is mitosis specific and hence has attracted attention as a target for mitosis-directed anticancer agents. KSP is a homotetrameric kinesin motor protein whose antiparallel microtubule-sliding function is required for separation of the two centrosomes in early mitosis. Since centrosome separation is essential for bipolar spindle formation, inhibition of KSP results in mitotic arrest due to formation of a microtubule array with only one pole. A number of KSP inhibitors have been described and some have progressed to clinical trials.^{21–23}

In-vitro and xenograft models

What is the evidence that inhibition of mitosis is the mechanism of action of MTAs and what are the models in which such data have been gathered? The evidence comes exclusively from *in vitro* cell culture and xenograft models.^{24,25} Primary tumor cells undergo selection to establish consistent cell lines with extended replicative capacity that can be used for *in vitro* culture and *in vivo* models. During immortalization the doubling time of primary

tumor cells significantly decreases; *in vitro*, doubling times are usually less than 24 h. For example, in the 60 cell lines of the NCI anticancer drug screen, the median doubling time is 33.1 h.²⁶ In such cell lines, the consequence of adding an MTA is mitotic arrest, a phenomenon that presents as G₂–M arrest usually within 24 h of drug administration (Figure 3a,b). *In vitro*, this mitotic arrest is followed by cell death, often detected by some measure of apoptosis such as 'ladder' formation upon DNA electrophoresis, a hallmark of DNA degradation.²⁷

Similar results are observed in xenograft models, the majority of which have been streamlined to have short doubling times that allow investigators to complete experiments in 'real time'. For example, in human tumor xenograft models doubling times range from 1.2 days to 12.2 days.²⁸ In the experiment depicted in Figure 3c,d, 24 h after the administration of 2-methoxyestradiol, an MTA advanced as an antiangiogenic therapy, one can see the drug's principal effect, microtubule disruption, with nearly half the cells arrested in mitosis. Such a marked mitotic arrest would never be observed in a patient. As might be expected, the consequence of

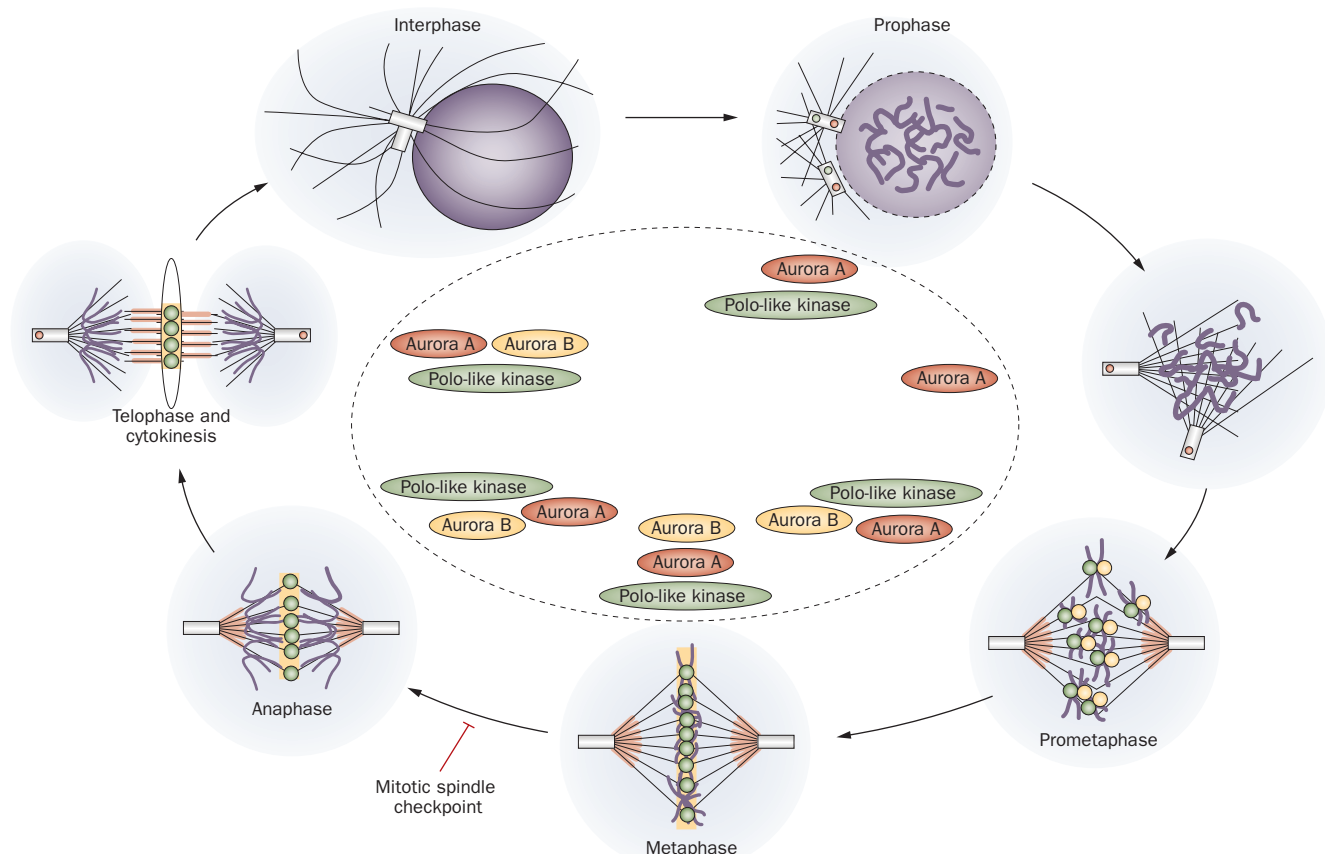


Figure 2 | The localization of aurora kinase A (red), aurora kinase B (orange), and polo-like kinases (green) during the different stages of mitosis. Permission obtained from Nature Publishing Group © Jackson, J. R. et al. *Nat. Rev. Cancer* 7, 107–117 (2007).

the rapid doubling times of *in vitro* and xenograft models is a vulnerability to a mitotic poison, a vulnerability that may not occur in the slower-growing tumors in patients.

Tumor doubling times

It is fortunate that the doubling times of human tumors are far greater than that of *in vitro* and xenograft models. Supplementary Tables 2–9 summarize data from studies that have examined tumor-doubling times using either radiographic measurements or serum tumor marker levels. The median doubling time for all tumors in this extensive but not exhaustive compilation is 147 days.

Studies that have looked at [³H]thymidine ([³H]TdR) labeling or flash doubling times have found values consistent with the median tumor doubling times summarized above.^{29,30} The basis of the percent-label technique is that cells actively synthesizing DNA incorporate [³H]TdR. Shortly after flash labeling with the [³H]TdR radiotracer, a specimen is removed, fixed, stained, and an autoradiograph prepared. For each sample the proportion of nucleated cells that have incorporated the label into their nuclei is determined. This labeling index represents the ratio of labeled cells or cells in DNA

synthesis to total cells of a defined morphologic type—such as tumor cells. Thus the flash labeling index (the proportion of cells in a population that incorporates [³H]TdR at a point in time) is a measure of proliferative activity. With doubling times of 30–60 days or longer one would not expect to find more than a small percentage of cells undergoing DNA synthesis at any one time.

Furthermore, analyses of the cell cycle have shown the amount (and fraction) of time spent in mitosis (ranging from 0.5 h to 2.5 h) is only a small percentage of the total doubling time,³¹ indicating that the longer doubling times are not a consequence of inordinately long mitosis, an observation that is largely self evident given that arrest in mitosis is intolerable to cells.^{32,33} By inference, one might argue that since a prolonged G₁ lasting weeks to months is also unlikely, cells might actually be spending a substantial fraction of their time out of the cell cycle in G₀. We recognize that the latter point can be debated but, regardless, it is clear that the fraction of time spent in mitosis and in turn the fraction of cells expressing the desired ‘mitotic-specific target’ are both small. Furthermore, when examining patient tumors, mitotic figures are rarely present

and are also not observed if the tumor is examined shortly after treatment with an MTA (Figure 4). In samples from breast cancer patients treated with ixabepilone (an epothilone analog that, similar to paclitaxel, stabilizes microtubules and causes mitotic arrest *in vitro* and in xenograft models),³⁴ despite extensive microtubule stabilization evidenced by increased staining with an antibody that recognizes stable microtubules (increased acetylated α -tubulin), there are only rare mitotic figures seen (Figure 4). This stands in stark contrast to the results in xenografts (Figure 3b) and underscores the longer doubling times of tumors in humans.

Myeloid kinetics

Not unexpectedly, the clinical development of mitosis-specific inhibitors encountered myelosuppression as the dose-limiting toxic effect. This toxic effect was likely mediated by target engagement in bone marrow precursors that, because of their constant turnover, are actively undergoing mitosis; neutrophil cellularity and kinetics underscore the mitotic activity of the normal bone marrow (Supplementary Table 10 online). Remarkably, at any one time 28% of marrow

neutrophils are in mitosis and the total marrow neutrophil production amounts to 0.85×10^9 cells/kg per day— 5.95×10^{10} cells/day in a 70 kg individual.³⁵ The latter means that the normal marrow output in 1 day is an amount similar to the number of cancer cells in seven to eight tumors of 2 cm in diameter, based on the estimation that a 1 cm³ tumor contains a billion cells. Given this, it is not surprising that neutropenia emerged as the dose-limiting toxic effect of mitosis-specific inhibitors.^{36,37}

Restricted expression

It is self evident that for a targeted therapy to be effective, the target must be present. For the mitotic kinases, the target is present exclusively in M-phase and occasionally in the G₂ phase of the cell cycle.³⁸ Notably, they are not expressed or are undetectable in S phase, G₁ phase and, by inference, in G₀ phase. In an *in vitro* or a xenograft model with a doubling time of 24–48 h, a large fraction of cells will prove vulnerable to a therapy that targets a mitosis-specific protein. It is not surprising that in a patient with a tumor doubling time of 30–60 days or greater and an S-phase fraction of at most a few percent, only a small, insignificant fraction of cells will be vulnerable to a drug aimed at a target that is expressed only transiently. As an example, the activity of the PLK-inhibitor BI 2536 (Boehringer Ingelheim, Bracknell, UK) in preclinical models has only been observed with frequent administration—twice weekly had improved efficacy compared with weekly—reflecting that even in models with rapid doubling times, a fraction of cells not in G₂–M phase are refractory to therapy and can repopulate the rapidly dividing tumor mass unless frequent administration is used.³⁹

Microtubules—intracellular traffic

In cancer, the focus on microtubules has often been on their role in mitosis; however, the microtubule cytoskeleton is important in intracellular trafficking. Supplementary Table 11 online presents a partial list of proteins that either associate with or traffic on microtubules and that are considered crucial to most cells, but especially cancer cells—the list includes many crucial oncoproteins such as p53,⁴⁰ BRCA1,^{41,42} Rb,⁴³ and androgen receptor.⁴⁴ For these proteins, an association with microtubules has been demonstrated by immunohistochemistry or in immunoprecipitation or pull-down experiments. Trafficking on microtubules has been inferred from an effect of treatment

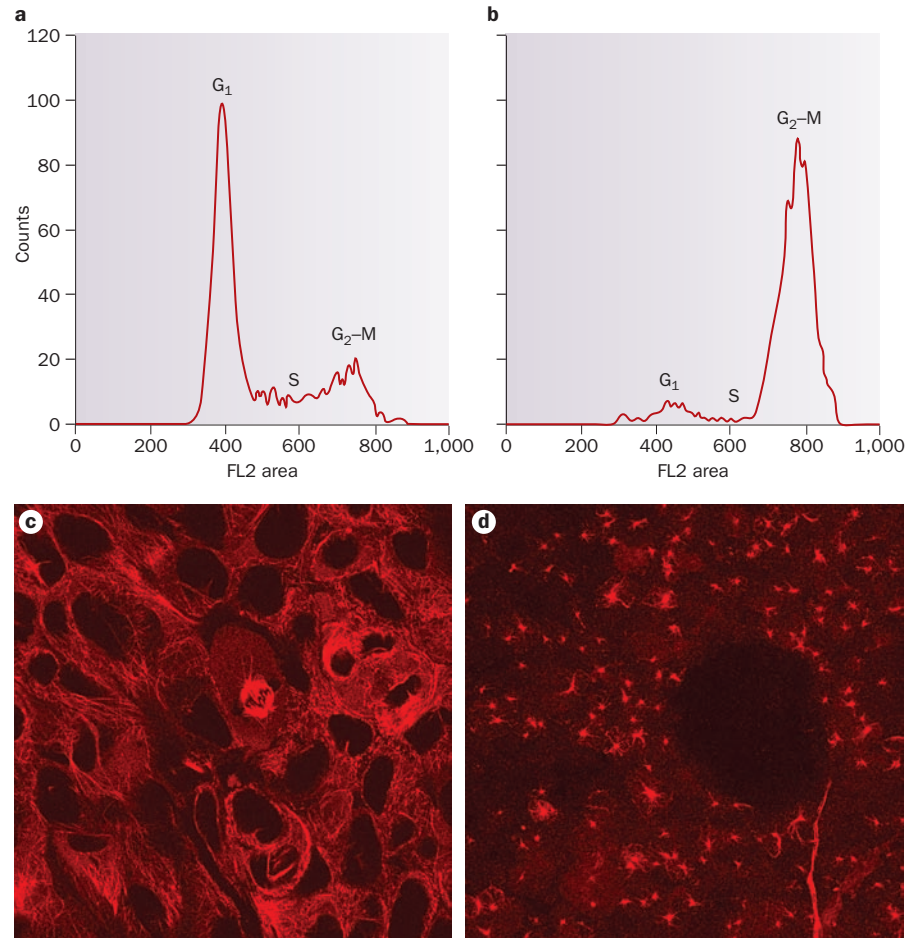


Figure 3 | Microtubule-targeting agents induce mitotic arrest and abnormal spindle in preclinical models. Paclitaxel treatment causes G₂–M arrest in A2780 (1A9) cells. **a** | Cells without drug and **b** | cells treated with 100 nM of paclitaxel for 24 h. Treatment of the MDA-MB231 orthotopic model with the microtubule-targeting agent 2-methoxyestradiol causes the display of abnormal mitotic spindles (asters). **c** | Control cells and **d** | cells 24 h after treatment with 2-methoxyestradiol and stained with anti-tubulin. Permission for parts c and d obtained from Elsevier © Mabeesh, N. J. *et al. Cancer Cell* 3, 363–375 (2003).

with an MTA or overexpression of dynamin, a protein that interferes with dynein-mediated trafficking.⁴³ We would argue that interfering with the ability of these crucial proteins to traffic on microtubules would be disruptive to a cancer cell and that such a disruption could lead to cell death.

Mitosis-independent toxicity

Can cell death independent of mitosis occur after administration of an MTA?⁴⁵ We believe the answer is yes, as evidenced by the fact that MTAs have major effects independent of mitosis. These effects include alterations of cell signaling and trafficking,^{46–48} including reactive oxygen species (ROS) signaling,⁴⁹ slowing of cell-cycle progression,⁵⁰ alterations of cell migration and invasiveness,⁵¹ and vascular disruption.⁵² Among these effects, vascular disruption characterized by rapid changes in the tumor

vasculature leading to infarction and necrosis has received the most attention. Reports of vascular disruption with low concentrations of some MTAs, such as the combretastatins, led investigators to refer to this class of MTAs as vascular-disrupting agents (VDAs). Although the mechanism of action of VDAs is not fully understood, exposure of endothelial cells to MTAs *in vitro* leads to numerous changes including the disruption of cell–cell adhesions, malformations of focal adhesions, induced actinomyosin contractility, accumulation of actin stress fibers, and phosphorylation of myosin light chain.⁵³ *In vivo*, the administration of VDAs is followed by rapid collapse and increased permeability of the established tumor with the center of the tumor more susceptible than the outer rim.⁵⁴ This rapid antivascular effect cannot be explained by inhibition of mitosis but rather by the morphological and

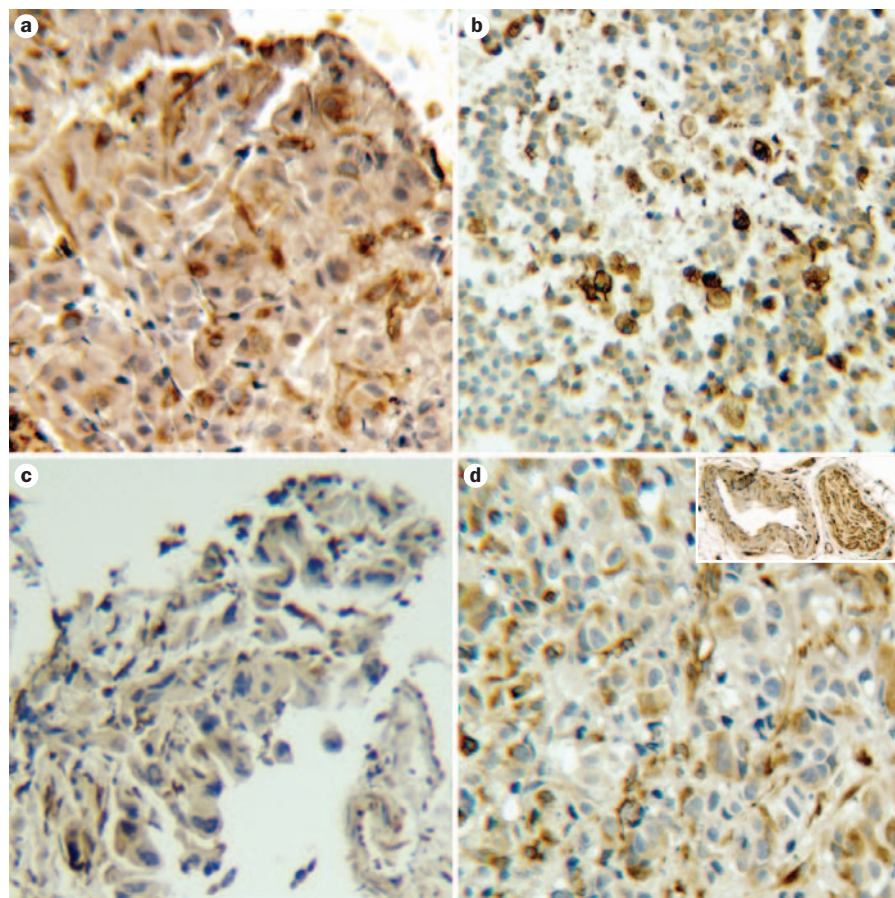


Figure 4 | Biopsies obtained from four patients with metastatic breast cancer after treatment with the microtubule-stabilizing agent, ixabepilone (6 mg/m² per day for 5 days). The biopsies show high levels of acetylated α -tubulin, reflecting target engagement and microtubule stabilization, and the absence of any mitotic figures. Biopsies were obtained on the second day of the second cycle approximately 24 h after the day 1 dose. **a** | Widespread occurrence of microtubule stabilization and the absence of cells arrested in mitosis. **b** | Staining reflecting microtubule stabilization is seen in cells, but not in the interstitium. **c** | Diffuse cytoplasmic staining is seen, with interphase nuclei that lack microtubules clearly lacking staining. **d** | The inset shows evidence of microtubule stabilization in a blood vessel and in a nerve—although the stabilization seen in the nerve can also be seen before the administration of ixabepilone since microtubules in nerves are very stable under normal circumstances and can stain with antibodies that detect stable microtubules, such as the anti-acetylated α -tubulin antibody used here. Magnification: 100–400-fold.

functional changes that occur in endothelial cells. These changes are due to the disruption of interphase microtubule function.

Evidence that MTAs can induce cell death independent of mitosis can also be found in the primary toxicity of MTAs—neurotoxicity—that is seen in as many as 70–90% of patients treated with MTAs.^{55–57} Peripheral neuropathy secondary to MTAs is usually bilateral, and preferentially involves sensory neurons over thick myelinated nerves. It commonly affects the distal lower limbs first and later involves the distal upper extremities in what is often described as a ‘stocking–glove distribution’, suggesting the longer the nerve is, the more vulnerable it is to damage.⁵⁸ This selectivity for long neurons argues that microtubule-dependent transport

from the neural cell body down the axon is being compromised. Although a limited number of studies have biopsied neurons, fiber loss, axonal atrophy and secondary demyelination have been observed following paclitaxel treatment.⁵⁹ Since neuronal cells rarely, if ever, divide in adults, in this instance MTAs have obviously caused the death of non-dividing cells.⁶⁰

Limited clinical activity

The development of mitosis-specific inhibitors as anticancer agents has come at a cost that now likely exceeds \$10 billion (Box 1). Their poor activity is summarized in Supplementary Table 12 online. Infrequent clinical responses have been observed in cases where a kinase other than a mitotic

kinase has emerged as a target. With tozasertib, for example, activity has been reported in patients with imatinib-resistant chronic myeloid leukemia harboring the Thr315Ile mutation, a mutant form of BCR-ABL, which tozasertib binds to with high affinity ($K_d = 5$ nM).⁶¹ ‘Activity characterized as stable disease’ has also been reported; for example, XL228 (Exelixis, San Francisco, CA, USA), a multitargeted protein kinase inhibitor targeting not only aurora A, but also inhibiting insulin-like growth factor 1 receptor, fibroblast growth factor receptors 1–3, ABL, ALK, and the SRC family of kinases, led to stable disease in 32% of cases in the escalating portion of the phase I study. In the same trial, stable disease was shown in 41% of cases treated with the maximum-tolerated dose including five out of nine (56%) patients with non-small-cell lung cancer.⁶² Given the large investment needed to develop a drug, it is not surprising that some have tried to rescue these drugs from their ultimate fate by assigning ‘stable disease’ as a measure of activity. While we doubt any ‘targeted agent’ can ‘arrest’ a cancer cell for weeks to months in any phase of the cell cycle other than G₀, it is doubtful that arresting a cell in mitosis for anything other than a brief period can ever happen. An arrest in mitosis sufficient to qualify as a meaningful ‘stable disease’—several months—will never happen. There is a reason cells find mitotic arrest intolerable: transcription from packed chromosomes is impossible.⁶³

Conclusions

Few targets can match the appeal the mitotic kinases have had. Scientifically exciting, involved in a crucial cellular function, and eminently ‘druggable’, the temptation was too great. The expectation was that they would be as good as, if not better than, MTAs but without neurotoxicity. The disappointing clinical results—a lack of meaningful activity but with marked myelosuppression—indicate the drugs taken to the clinic were well designed and the target was a valid one for rapidly dividing cells. The high affinity of the drugs synthesized and developed clinically, many of which have K_i values in the nM or sub-nM range, are a testament to the skill of the scientists who have developed these compounds. The emergence of neutropenia as a toxic effect of this class of agents is strong evidence that the pharmacodynamics were excellent and that their pharmacokinetics were likewise favorable. Although the clinical data are limited, there are extensive animal data that, together with

Box 1 | The cost of drug development

A report by DiMasi and colleagues⁶⁷ estimated that for drugs entering human clinical trials for the first time between 1989 and 2002, the cost to develop a new drug was \$868 million. Adams and Brantner,⁶⁸ using the published cost estimates of DiMasi *et al.*,⁶⁷ along with information on success rates and durations from a publicly available data set estimated costs to “vary from around \$500 million to more than \$2,000 million, depending on the therapy or the developing firm”. For cancer drugs, the estimate was \$1,042 million; with available data allowing an estimate for breast cancer of \$610 million. With at least 25 mitosis-specific agents (16 aurora kinase inhibitors, five polo-like kinase inhibitors and four kinesin-spindle protein inhibitors [Supplementary Table 12 online]) having reported results from clinical trials and more under development or yet to be reported, even an expenditure of only a fraction of this estimate per agent would mean that billions have been spent on developing this class of compounds. The \$10 billion is an estimate based on these numbers; the actual number cannot be known with certainty and could be somewhat more or somewhat less than this, but nevertheless it is a very substantial expenditure.

the expected neutropenia in human subjects, suggest their lack of activity has not been a pharmacokinetic problem. However, as the data presented in this Perspective demonstrate, the majority of human tumors do not divide rapidly enough to be susceptible to such drugs and, more importantly, they divide much slower than vulnerable marrow elements. We now have data that validate a paradigm for how MTAs kill cancer cells in patients—principally by interfering with interphase microtubule functions, for example intracellular trafficking—not by disrupting mitosis.

The novel paradigm we propose for the mechanism of action of MTAs is concordant with established and emerging data demonstrating that therapies we might have previously considered to be cell-cycle specific in fact have diverse targets and activity in phases of the cell cycle other than those originally envisioned. The anthracyclines, for example, were originally envisioned as DNA-interacting agents, but research subsequently uncovered multiple cellular targets and effects including the ability to create iron-mediated free oxygen radicals that damage cellular components including membranes, intercalate between base pairs interfering with DNA and RNA metabolism, covalently bind to DNA and inhibit topoisomerase II.⁶⁴ This complex mechanism of action likely contributes to their efficacy, underlies the diverse mechanisms of resistance that can emerge in a cell treated with these agents and helps explain its toxicity profile, including cardiotoxicity, an adverse event that, like neurotoxicity, targets cells that either do not or only rarely divide. Similarly, the camptothecins that are known for their ability to specifically bind and inhibit a precise step of the Top1 reaction—binding only to the complex formed by Top1 when it cleaves DNA (that is, the cleavage complex)—have

diverse cellular consequences throughout the cell cycle with Top1 cleavage complexes leading to both replication-mediated and transcription-mediated DNA damage.⁶⁵ Finally, antimetabolites thought initially to inhibit DNA synthesis by incorporation into DNA are now known to inhibit critical enzymes and target both RNA and DNA. For example, 5-fluorouracil inhibits thymidylate synthase and more importantly incorporates into RNA, altering mRNA expression, while inhibiting mRNA splicing and pre-ribosomal RNA processing, and interfering with transfer RNA modification. Indeed, in humans treated with 5-fluorouracil, examination of tumors and normal mucosa has shown that 5-fluorouracil is incorporated more rapidly into RNA than into DNA and retained longer in RNA than in DNA, reaching levels in RNA that are 8 to 30-fold higher than those in DNA.⁶⁶

Given the widespread use of MTAs in a broad range of malignancies, especially in combinations, and the likelihood they will continue as essential components of our armamentarium, we should begin to think of MTAs as highly specific ‘targeted agents’ that owe activity in more slowly dividing cells, such as the majority of tumors in humans, to interference with interphase functions such as microtubule trafficking and only manifest a primarily antimitotic activity in rapidly dividing cells such as the normal bone marrow, animal models or *in vitro* cell culture.

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- Jordan, M. A. & Kamath, K. How do microtubule-targeted drugs work? An overview. *Curr. Cancer Drug Targets* **7**, 730–742 (2007).
- Brinkley, B. R. Microtubule organizing centers. *Annu. Rev. Cell Biol.* **1**, 145–172 (1985).
- Terada, S. & Hirokawa, N. Moving on to the cargo problem of microtubule-dependent motors in neurons. *Curr. Opin. Neurobiol.* **10**, 566–573 (2000).
- Marshall, W. F. Basal bodies platforms for building cilia. *Curr. Top. Dev. Biol.* **85**, 1–22 (2008).
- Bergen, L. G. & Borisy, G. G. Head-to-tail polymerization of microtubules *in vitro*. Electron microscope analysis of seeded assembly. *J. Cell Biol.* **84**, 141–150 (1980).
- Bergen, L. G., Kuriyama, R. & Borisy, G. G. Polarity of microtubules nucleated by centrosomes and chromosomes of Chinese hamster ovary cells *in vitro*. *J. Cell Biol.* **84**, 151–159 (1980).
- Hirokawa, N., Noda, Y., Tanaka, Y. & Niwa, S. Kinesin superfamily motor proteins and intracellular transport. *Nat. Rev. Mol. Cell Biol.* **10**, 682–696 (2009).
- Kardon, J. R. & Vale, R. D. Regulators of the cytoplasmic dynein motor. *Nat. Rev. Mol. Cell Biol.* **10**, 854–865 (2009).
- Ito, D. & Matsumoto, T. Molecular mechanisms and function of the spindle checkpoint, a guardian of the chromosome stability. *Adv. Exp. Med. Biol.* **676**, 15–26 (2010).
- Schmit, T. L. & Ahmad, N. Regulation of mitosis via mitotic kinases: new opportunities for cancer management. *Mol. Cancer Ther.* **6**, 1920–1931 (2007).
- Colombo, R. & Moll, J. Target validation and biomarker identification in oncology: the example of aurora kinases. *Mol. Diagn. Ther.* **12**, 71–76 (2008).
- Chopra, P., Sethi, G., Dastidar, S. G. & Ray, A. Polo-like kinase inhibitors: an emerging opportunity for cancer therapeutics. *Expert Opin. Investig. Drugs* **19**, 27–43 (2010).
- Taylor, S. & Peters, J. M. Polo and Aurora kinases: lessons derived from chemical biology. *Curr. Opin. Cell Biol.* **20**, 77–84 (2008).
- Nakai, R. *et al.* K858, a novel inhibitor of mitotic kinesin Eg5 and antitumor agent, induces cell death in cancer cells. *Cancer Res.* **69**, 3901–3909 (2009).
- Harrison, M. R., Holen, K. D. & Liu, G. Beyond taxanes: a review of novel agents that target mitotic tubulin and microtubules, kinases, and kinesins. *Clin. Adv. Hematol. Oncol.* **7**, 54–64 (2009).
- Jackson, J. R., Patrick, D. R., Dar, M. M. & Huang, P. S. Targeted anti-mitotic therapies: can we improve on tubulin agents? *Nat. Rev. Cancer* **7**, 107–117 (2007).
- Duterte, S., Descamps, S. & Prigent, C. On the role of aurora-A in centrosome function. *Oncogene* **21**, 6175–6183 (2002).
- Sasai, K. *et al.* Aurora-C kinase is a novel chromosomal passenger protein that can complement Aurora-B kinase function in mitotic cells. *Cell Motil. Cytoskeleton* **59**, 249–263 (2004).
- van de Weert, B. C. & Medema, R. H. Polo-like kinases: a team in control of the division. *Cell Cycle* **5**, 853–864 (2006).
- Kishi, K., van Vugt, M. A., Okamoto, K., Hayashi, Y. & Yaffe, M. B. Functional dynamics of Polo-like kinase 1 at the centrosome. *Mol. Cell Biol.* **29**, 3134–3150 (2009).
- Huszar, D., Theoclitou, M. E., Skolnik, J. & Herbst, R. Kinesin motor proteins as targets

- for cancer therapy. *Cancer Metastasis Rev.* **28**, 197–208 (2009).
22. Sarli, V. & Giannis, A. Targeting the kinesin spindle protein: basic principles and clinical implications. *Clin. Cancer Res.* **14**, 7583–7587 (2008).
 23. Zhang, Y. & Xu, W. Progress on kinesin spindle protein inhibitors as anti-cancer agents. *Anticancer Agents Med. Chem.* **8**, 698–704 (2008).
 24. Milas, L. *et al.* Kinetics of mitotic arrest and apoptosis in murine mammary and ovarian tumors treated with taxol. *Cancer Chemother. Pharmacol.* **35**, 297–303 (1995).
 25. Horton, J. K., Houghton, P. J. & Houghton, J. A. Relationships between tumor responsiveness, vincristine pharmacokinetics and arrest of mitosis in human tumor xenografts. *Biochem. Pharmacol.* **37**, 3995–4000 (1988).
 26. NCI Developmental Therapeutics Program Discovery Services [online], http://dtp.nci.nih.gov/docs/misc/common_files/cell_list.html (2010).
 27. Bekier, M. E., Fischbach, R., Lee, J. & Taylor, W. R. Length of mitotic arrest induced by microtubule-stabilizing drugs determines cell death after mitotic exit. *Mol. Cancer Ther.* **8**, 1646–1654 (2009).
 28. Corbett, T. *et al.* in *Anticancer Drug Development Guide: Preclinical Screening, Clinical Trials, and Approval* (eds Teicher, B. A. & Andrews, P. A.) 99–125 (Humana Press, New Jersey, 2004). [Series Ed. Teicher, B. A. *Cancer Drug Discovery and Development*].
 29. Tubiana, M. & Malaise, E. in *Scientific Foundations of Oncology* (eds Symington, T. & Carter, R. L.) 126–136 (William Heinemann Medical Books, London, 1976).
 30. Livingston, R. B., Ambus, U., George, S. L., Freireich, E. J. & Hart, J. S. *In vitro* determination of thymidine-³H labeling index in human solid tumors. *Cancer Res.* **34**, 1376–1380 (1974).
 31. Baserga, R. The relationship of the cell cycle to tumor growth and control of cell division: a review. *Cancer Res.* **25**, 581–595 (1965).
 32. Smith, J. A. & Martin, L. Do cells cycle? *Proc. Natl Acad. Sci. USA* **70**, 1263–1267 (1973).
 33. Rixe, O. & Fojo, T. Is cell death a critical end point for anticancer therapies or is cytostasis sufficient? *Clin. Cancer Res.* **13**, 7280–7287 (2007).
 34. Denduluri, N. *et al.* Phase II trial of ixabepilone, an epothilone B analog, in patients with metastatic breast cancer previously untreated with taxanes. *J. Clin. Oncol.* **25**, 3421–3427 (2007).
 35. Dancey, J. T., Deubelbeiss, K. A., Harker, L. A. & Finch, C. A. Neutrophil kinetics in man. *J. Clin. Invest.* **58**, 705–715 (1976).
 36. Steeghs, N. *et al.* Phase I pharmacokinetic and pharmacodynamic study of the aurora kinase inhibitor danusertib in patients with advanced or metastatic solid tumors. *J. Clin. Oncol.* **27**, 5094–5101 (2009).
 37. Moss, K. *et al.* Phase I dose escalation and pharmacokinetic study of BI 2536, a novel Polo-like kinase 1 inhibitor, in patients with advanced solid tumors. *J. Clin. Oncol.* **26**, 5511–5517 (2008).
 38. Crosio, C. *et al.* Mitotic phosphorylation of histone H3: spatio-temporal regulation by mammalian Aurora kinases. *Mol. Cell. Biol.* **22**, 874–885 (2002).
 39. Steegmaier, M. *et al.* BI 2536, a potent and selective inhibitor of polo-like kinase 1, inhibits tumor growth *in vivo*. *Curr. Biol.* **17**, 316–322 (2007).
 40. Giannakakou, P. *et al.* p53 is associated with cellular microtubules and is transported to the nucleus by dynein. *Nat. Cell. Biol.* **2**, 709–717 (2000).
 41. Hsu, L. C. & White, R. L. BRCA1 is associated with the centrosome during mitosis. *Proc. Natl Acad. Sci. USA* **95**, 12983–12988 (1998).
 42. Lotti, L. V. *et al.* Subcellular localization of the BRCA1 gene product in mitotic cells. *Genes Chromosomes Cancer* **35**, 193–203 (2002).
 43. Roth, D. M., Moseley, G. W., Glover, D., Pouton, C. W. & Jans, D. A. A microtubule-facilitated nuclear import pathway for cancer regulatory proteins. *Traffic* **8**, 673–686 (2007).
 44. Zhu, M. L. *et al.* Tubulin-targeting chemotherapy impairs androgen receptor activity in prostate cancer. *Cancer Res.* **70**, 7992–8002 (2010).
 45. Gundersen, G. G. & Cook, T. A. Microtubules and signal transduction. *Curr. Opin. Cell Biol.* **11**, 81–94 (1999).
 46. Ganansia-Leymarie, V., Bischoff, P., Bergerat, J. P. & Holl, V. Signal transduction pathways of taxanes-induced apoptosis. *Curr. Med. Chem. Anticancer Agents* **3**, 291–306 (2003).
 47. Herbst, R. S. & Khuri, F. R. Mode of action of docetaxel—a basis for combination with novel anticancer agents. *Cancer Treat. Rev.* **29**, 407–415 (2003).
 48. Etienne-Manneville, S. From signaling pathways to microtubule dynamics: the key players. *Curr. Opin. Cell Biol.* **22**, 104–111 (2010).
 49. Jang, H. J. *et al.* Taxol induces oxidative neuronal cell death by enhancing the activity of NADPH oxidase in mouse cortical cultures. *Neurosci. Lett.* **443**, 17–22 (2008).
 50. Pasquier, E. *et al.* Antiangiogenic activity of paclitaxel is associated with its cytostatic effect, mediated by the initiation but not completion of a mitochondrial apoptotic signaling pathway. *Mol. Cancer Ther.* **3**, 1301–1310 (2004).
 51. Tran, T. A. *et al.* Non-anti-mitotic concentrations of taxol reduce breast cancer cell invasiveness. *Biochem. Biophys. Res. Commun.* **379**, 304–308 (2009).
 52. Kanthou, C. & Tozer, G. M. Tumour targeting by microtubule-depolymerizing vascular disrupting agents. *Expert Opin. Ther. Targets* **11**, 1443–1457 (2007).
 53. Pasquier, E., André, N. & Braguer, D. Targeting microtubules to inhibit angiogenesis and disrupt tumour vasculature: implications for cancer treatment. *Curr. Cancer Drug Targets* **7**, 566–581 (2007).
 54. Dark, G. G. *et al.* Combretastatin A-4, an agent that displays potent and selective toxicity toward tumor vasculature. *Cancer Res.* **57**, 1829–1834 (1997).
 55. Forsyth, P. A. *et al.* Prospective study of paclitaxel-induced peripheral neuropathy with quantitative sensory testing. *J. Neurooncol.* **35**, 47–53 (1997).
 56. Postma, T. J., Vermorken, J. B., Liefing, A. J., Pinedo, H. M. & Heimans, J. J. Paclitaxel-induced neuropathy. *Ann. Oncol.* **6**, 489–494 (1995).
 57. DeAngelis, L. M., Gnecco, C., Taylor, L. & Warrell, R. P. Jr. Evolution of neuropathy and myopathy during intensive vincristine/corticosteroid chemotherapy for non-Hodgkin's lymphoma. *Cancer* **67**, 2241–2246 (1991).
 58. Mielke, S., Sparreboom, A. & Moss, K. Peripheral neuropathy: a persisting challenge in paclitaxel-based regimens. *Eur. J. Cancer* **42**, 24–30 (2006).
 59. Sahenk, Z., Barohn, R., New, P. & Mendell, J. R. Taxol neuropathy. Electrodiagnostic and sural nerve biopsy findings. *Arch. Neurol.* **51**, 726–729 (1994).
 60. Argyriou, A. A., Koltzenburg, M., Polychronopoulos, P., Papapetropoulos, S. & Kalofonos, H. P. Peripheral nerve damage associated with administration of taxanes in patients with cancer. *Crit. Rev. Oncol. Hematol.* **66**, 218–228 (2008).
 61. Giles, F. J. *et al.* MK-0457, a novel kinase inhibitor, is active in patients with chronic myeloid leukemia or acute lymphocytic leukemia with the T3151 BCR-ABL mutation. *Blood* **109**, 500–502 (2007).
 62. Smith, D. C. *et al.* A phase I study of XL228, a multitargeted protein kinase inhibitor, in patients (pts) with solid tumors or multiple myeloma [abstract]. *J. Clin. Oncol.* **28** (15 Suppl.), a3105 (2010).
 63. Andersson, K., Mahr, R., Björkroth, B. & Daneholt, B. Rapid reformation of the thick chromosome fiber upon completion of RNA synthesis at the Balbiani ring genes in *Chironomus tentans*. *Chromosoma* **87**, 33–48 (1982).
 64. Carvalho, C. *et al.* Doxorubicin: the good, the bad and the ugly effect. *Curr. Med. Chem.* **16**, 3267–3285 (2009).
 65. Pommier, Y. Camptothecins and topoisomerase I: a foot in the door. Targeting the genome beyond topoisomerase I with camptothecins and novel anticancer drugs: importance of DNA replication, repair and cell cycle checkpoints. *Curr. Med. Chem. Anticancer Agents* **4**, 429–434 (2004).
 66. Noordhuis, P., Holwerda, U., Van Laar, J. A., Van der Wilt, C. L. & Peters, G. J. A non-radioactive sensitive assay to measure 5-fluorouracil incorporation into DNA of solid tumors. *Nucleosides Nucleotides Nucleic Acids* **23**, 1481–1484 (2004).
 67. DiMasi, J. A., Hansen, R. W. & Grabowski, H. G. The price of innovation: new estimates of drug development costs. *J. Health Econ.* **22**, 151–185 (2003).
 68. Adams, C. P. & Brantner, V. V. Estimating the cost of new drug development: is it really 802 million dollars? *Health Aff. (Millwood)* **25**, 420–428 (2006).

Acknowledgments

The authors would like to thank S. X.Yung for immunohistochemical staining of breast cancer samples. This work was supported in part by the Intramural Research Program of the Eunice Kennedy Shriver National Institute of Child Health and Human Development.

Author contributions

All authors contributed to researching the data for the article. E. Komlodi-Pasztor, D. Sackett and T. Fojo made a substantial contribution to discussion of the content, writing the article and preparing it for submission.

Supplementary information is linked to the online version of the paper at www.nature.com/nrclinonc

CORRECTION**OPINION: Mitosis is not a key target of microtubule agents in patient tumors**

Edina Komlodi-Pasztor, Dan Sackett, Julia Wilkerson and Tito Fojo

In the version of this article initially published online, the legend for Figure 3 should have contained the sentence 'Permission for parts c and d obtained from Elsevier © Mabweesh, N. J. *et al. Cancer Cell* 3, 363–375 (2003)'. This has now been corrected for the print, HTML and PDF versions of the article.